

A FIRST-PRINCIPLES CONSTRUCTION OF THETA-EXPECTATIONS FROM DETERMINISTIC BILLIARDS

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ABSTRACT. We derive a nonlinear HJB limit and the associated theta-expectation semigroup from deterministic finite-horizon dispersing billiards with a finite-response mechanical port. The singular-billiard response package is imported as a named theorem interface from the companion response-theory paper: primitive spectral/resolvent control, the suspension Laplace contour, moving-singularity response, trace insertion into resolvent words, uniform multi-response bounds, and the finite regularity-loss budget have already been derived from primitive billiard geometry. Using only that exported interface and the deterministic finite-response action, we construct the Green-Kubo diffusion tensor, the effective Hamiltonian, the perturbed-test corrector hierarchy, and the viscosity limit. The resulting Hamiltonian may be nonconvex in the gradient, and the induced theta-expectation is not generally subadditive.

1. INTRODUCTION

This paper is the second article in a three-paper sequence. Its task is not to reprove singular-billiard response theory. Its task is to use the companion response theorem, specifically Paper 1, *First-principles response package*, as the already verified dynamical engine and then prove the deterministic homogenization and nonlinear expectation theorem.

First-principles interface. The phrase first principles means the following dependency chain:

$$(1) \quad \begin{aligned} & \text{primitive billiard geometry and finite-response mechanics} \\ \implies & \text{ Paper 1 first-principles response package} \implies \text{deterministic HJB limit} \\ & \implies \text{theta-expectation semigroup.} \end{aligned}$$

BSDE, Girsanov, and payoff-dependent representation formulae are not used in this paper's derivation; they belong to the representation-calculus paper.

1.1. Theorem map.

Layer	Paper theorem	Proof role
Response input	Assumption 2.1	companion theorem imported, not reproved
Coefficients	Definition 3.2 and Proposition 3.3	deterministic transfer-resolvent coefficients
Limit theorem	Theorem 4.1	half-relaxed viscosity closure
Nonconvexity	Theorem 5.1	concrete port and short-time Hamiltonian defect
Semigroup	Definition 5.2 and Theorem 5.3	nonlinear expectation output

1.2. Provenance and scope. The singular-billiard estimates used here are imported only through the companion theorem interface in Assumption 2.1. The present paper cites the standard PDE and nonlinear-expectation technology it uses: viscosity solutions and comparison follow [3, 5], half-relaxed stability and monotone consistency follow [1], perturbed-test homogenization is in the line of [2, 4, 6], and the sublinear benchmark called a G -expectation follows the nonlinear expectation

terminology in [7]. These references supply the macroscopic language; the coefficients themselves are deterministic transfer-resolvent outputs from Paper 1 and the finite-response port.

The technical appendices record the finite-response micro-action ledger, the full contact-frozen corrector hierarchy, the four-scale residual identity, the half-relaxed viscosity passage, and the coefficient/nonconvexity checks supporting the proof below.

2. RESPONSE THEOREM IMPORTED FROM PAPER 1

Assumption 2.1 (Companion response theorem as imported theorem). The companion response-theory paper proves, from primitive finite-horizon dispersing billiard geometry and the compact finite-response port, the following package:

- (i) an a.e. singular hyperbolic collision map and suspension flow;
 - (ii) uniform anisotropic transfer spaces and reduced observable resolvents;
 - (iii) moving-singularity response formulae in the Banach bundle;
 - (iv) a finite regularity-loss budget for all coefficient derivatives used in the present HJB derivation.
- This item is a theorem of Paper 1, not a macroscopic assumption in the present paper. The exact imported interface is Paper 1, *First-principles response package*. The parts of that interface used below are:

Paper 1 source	Use in the present paper
Paper 1, <i>Primitive spectral and resolvent theorem</i>	spectral gap, invariant projector, zero-mean reduced observable resolvent, and Green-Kubo correlation control.
Paper 1, <i>Suspension Laplace contour and frequency partition</i>	continuous-time reduced resolvent and contour bounds for suspension correctors.
Paper 1, <i>Smooth deterministic response</i> and Paper 1, <i>Moving-singularity response theorem</i>	differentiability of invariant pairings, correctors, projectors, and coefficient functionals in the moving billiard fiber.
Paper 1, <i>Trace insertion into resolvent words</i> and Paper 1, <i>Uniform multi-response bounds</i>	bounded differentiated resolvent words, including moving-boundary trace insertions.
Paper 1, <i>Finite regularity-loss budget</i>	the eight-derivative primitive loss used by the HJB coefficient calculus.

3. FINITE-RESPONSE MECHANICS AND EFFECTIVE COEFFICIENTS

Definition 3.1 (Finite-response prelimit data). The slow variable is $x \in \mathbb{R}^m$ and the microscopic variable lies in the suspension billiard fiber selected by a finite-response parameter $\Theta(x, p)$. The mechanical port provides branchwise smooth read-outs:

$$(2) \quad B(x, p, z), \quad V(x, p, z), \quad \Theta(x, p),$$

where B is the centered slow drift after subtracting the instantaneous microcanonical mean and V is the action-density contribution. These read-outs are primitive finite-response objects; they are fixed before the effective Hamiltonian is defined.

Definition 3.2 (Effective coefficients). For each compact jet window, Paper 1, *Primitive spectral and resolvent theorem*, with the continuous-time contour control from Paper 1, *Suspension Laplace contour and frequency partition*, gives the zero-mean corrector

$$(3) \quad \chi_{x,p} = (-\mathcal{A}_{\Theta(x,p)}|_{\ker \Pi_{\Theta(x,p)}})^{-1} B(x, p, \cdot).$$

The diffusion tensor and Hamiltonian are deterministic pairings in the anisotropic observable bundle:

$$(4) \quad D(x, p) = \int_0^\infty \langle B(x, p, \cdot), B(x, p, \Phi_{\Theta(x,p)}^t \cdot) \rangle_{\Theta(x,p)} dt,$$

$$(5) \quad H(x, p) = \langle V(x, p, \cdot) \rangle_{\Theta(x,p)} + \text{the finite-response work and corrector contributions.}$$

The HJB operator is

$$(6) \quad \mathcal{H}(x, p, X) = -\operatorname{tr}(D(x, p)X) - H(x, p).$$

Proposition 3.3 (Coefficient structure). *On compact jet windows, D is symmetric and positive semidefinite, $H(x, 0) = 0$, and $\mathcal{H}$ is affine in the Hessian variable X . The dependence on the gradient p is inherited from the finite-response port and need not be convex. All derivatives used below are obtained from Paper 1, Smooth deterministic response, Paper 1, Uniform multi-response bounds, and the finite-response read-outs.*

Definition 3.4 (Corrector hierarchy and perturbed test). Let $\varphi \in C^3$ be a macroscopic test function and set $p = D\varphi(t, x)$ and $X = D^2\varphi(t, x)$. The first corrector is the zero-mean solution

$$(7) \quad -\mathcal{A}_{\Theta(x, p)}\chi_{x, p} = B(x, p, \cdot), \quad \Pi_{\Theta(x, p)}\chi_{x, p} = 0.$$

The second corrector $\psi_{x, p, X}$ is the zero-mean solution of the cell equation whose right-hand side is the centered second-order transport/action residual after inserting $\chi_{x, p}$. The perturbed test function is

$$(8) \quad \varphi^\varepsilon(t, x, z) = \varphi(t, x) + \varepsilon\chi_{x, p(t, x)}(z) + \varepsilon^2\psi_{x, p(t, x), X(t, x)}(z).$$

All correctors are transfer-resolvent outputs supplied by Paper 1, *Primitive spectral and resolvent theorem* and its suspension version Paper 1, *Suspension Laplace contour and frequency partition*.

Lemma 3.5 (Deterministic residual identity). *At a viscosity contact point, freeze the finite-response parameter at $\Theta(x, D\varphi)$ and insert Definition 3.4 into the deterministic prelimit Hamilton-Jacobi operator. Then*

$$(9) \quad \mathcal{L}^\varepsilon\varphi^\varepsilon = \partial_t\varphi + \mathcal{H}(x, D\varphi, D^2\varphi) + R_\varphi^\varepsilon,$$

where R_φ^ε is an anisotropic-duality residual satisfying, on every compact jet window,

$$(10) \quad |\langle R_\varphi^\varepsilon, \zeta \rangle| \leq \omega_M(\varepsilon)\|\zeta\|_{\mathcal{B}^\#}(1 + \|\varphi\|_{C^3}), \quad \omega_M(\varepsilon) \rightarrow 0.$$

The identity contains no uncanceled ε^{-1} term and no pointwise claim that a zero-mean microscopic observable is small at a maximum point.

Proof. All expansions are made after freezing the actual contact jet. Write

$$p_\varepsilon = D\varphi(t_\varepsilon, x_\varepsilon), \quad X_\varepsilon = D^2\varphi(t_\varepsilon, x_\varepsilon), \quad \theta_\varepsilon = \Theta(x_\varepsilon, p_\varepsilon).$$

The first corrector is evaluated in the fiber over θ_ε and satisfies

$$-\mathcal{A}_{\theta_\varepsilon}\chi_{x_\varepsilon, p_\varepsilon} = B(x_\varepsilon, p_\varepsilon, \cdot).$$

Consequently the scale- ε^{-1} terms in the branchwise prelimit operator cancel at the contact jet before any limiting passage is taken. This prevents an uncanceled $\varepsilon^{-1}(p_\varepsilon - D\varphi(t_0, x_0))$ error.

The order-one part contains the finite-response action read-out and the work created by the horizontal derivative of the first corrector. The physical gradient increment has the form

$$q_\varepsilon = D_x^{\text{fr}}\chi_{x_\varepsilon, p_\varepsilon} + X_\varepsilon^T\chi_{x_\varepsilon, p_\varepsilon},$$

where χ denotes the vector of first correctors for the slow coordinates. Paper 1, *Smooth deterministic response*, combined with Paper 1, *Uniform multi-response bounds*, identifies the invariant projection of this complete order-one observable with

$$\operatorname{tr}(D(x_\varepsilon, p_\varepsilon)X_\varepsilon) + H(x_\varepsilon, p_\varepsilon).$$

Because $\mathcal{H}(x, p, X) = -\operatorname{tr}(D(x, p)X) - H(x, p)$, this is exactly the coefficient contribution appearing in $\partial_t\varphi + \mathcal{H}(x, D\varphi, D^2\varphi)$ after moving the prelimit operator to the viscosity side. The centered order-one part is solved by the second cell equation defining $\psi_{x, p, X}$.

Every term not cancelled in these two steps has at least one small factor or a controlled contact-jet displacement. The list is finite: Taylor remainders of the smooth test, horizontal products involving

the normalized first and second correctors, finite-response parameter variation on the compact jet window, and moving-singularity boundary traces. Paper 1 supplies the product, response, and strip-tail bounds for these terms through Paper 1, *Trace insertion into resolvent words*, Paper 1, *Uniform multi-response bounds*, and Paper 1, *Finite regularity-loss budget*. Pairing with $\zeta \in \mathcal{B}^\#$ therefore gives the stated residual estimate with $\omega_M(\varepsilon) \rightarrow 0$. $\square$

Lemma 3.6 (Contact-frozen perturbed tests). *Let $\varphi \in C^3$ touch the upper half-relaxed limit strictly at (t_0, x_0) . Then, for ε small, there are prelimit contact points $(t_\varepsilon, x_\varepsilon, z_\varepsilon) \rightarrow (t_0, x_0, z_0)$ outside the zero-measure singular strata such that the perturbed test in Definition 3.4, frozen at $(t_\varepsilon, x_\varepsilon)$ and normalized by subtracting its microscopic value at that contact point, is an admissible upper viscosity-duality test for u^ε . The analogous lower-contact statement holds for lower half-relaxed limits.*

Proof. Make the macroscopic contact strict by adding a small positive bump and later letting the bump size go to zero. The half-relaxed definition then gives prelimit maximizing triples in every compact cylinder. The iterated grazing and multiple-collision strata have zero Liouville flux measure, and admissible density tests have summable high-strip tails, so the triples may be chosen in homogeneous branches. The first and second cell equations are solved after freezing the actual contact jet

$$p_\varepsilon = D\varphi(t_\varepsilon, x_\varepsilon), \quad X_\varepsilon = D^2\varphi(t_\varepsilon, x_\varepsilon).$$

Subtracting the microscopic gauge value at the contact point leaves all generator and horizontal derivatives unchanged but makes the contact equality valid for the normalized test. This is the point that replaces any invalid claim that a zero-mean corrector is pointwise small at a maximum. $\square$

Proposition 3.7 (Subsolution and supersolution residuals). *The upper half-relaxed limit of the deterministic prelimit solutions is a viscosity subsolution of the effective HJB equation, and the lower half-relaxed limit is a viscosity supersolution. In both cases the only microscopic input is Lemma 3.5; singular-billiard estimates enter only through the named Paper 1 interface in Assumption 2.1.*

Proof. For the subsolution inequality, let φ touch the upper relaxed limit from above. Use Lemma 3.6 to choose homogeneous prelimit contact points and the normalized contact-frozen perturbed test φ^ε . Applying the prelimit viscosity-duality subsolution inequality to this test and then using Lemma 3.5 gives

$$\partial_t \varphi + \mathcal{H}(x, D\varphi, D^2\varphi) \leq |\langle R_\varphi^\varepsilon, \zeta_\varepsilon \rangle|.$$

The residual bound is uniform on compact jet windows and bounded admissible density balls, hence the right-hand side tends to zero. This gives the subsolution inequality. The lower-contact construction in Lemma 3.6 is identical, and the viscosity-duality inequality is reversed; this gives the supersolution inequality. $\square$

Proposition 3.8 (Comparison modulus for the derived HJB). *On every compact jet window, the Hamiltonian derived in Definition 3.2 satisfies the Crandall-Ishii continuity modulus required for viscosity comparison. More precisely, if $X, Y \in \text{Sym}_m$ and $\lambda > 0$ satisfy*

$$(11) \quad -3\lambda \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \leq \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \leq 3\lambda \begin{pmatrix} I & -I \\ -I & I \end{pmatrix},$$

then there is a modulus ω_M such that

$$(12) \quad \mathcal{H}(y, p, Y) - \mathcal{H}(x, p, X) \leq \omega_M(|x - y|(1 + |p|) + \lambda|x - y|^2)$$

for all (x, y, p, X, Y) in that window. The modulus is a deterministic consequence of the finite-response saturation, Paper 1's smooth-response and multi-response bounds, and the Green-Kubo factorization of the diffusion tensor.

Proof. Paper 1, *Smooth deterministic response* and Paper 1, *Uniform multi-response bounds* give the required horizontal regularity of the resolvent outputs that define D and H . Finite-response compactness keeps the parameter selector, branch charts, invariant projectors, and port derivatives in one compact family on the chosen jet window. Hence

$$\|D(x, p) - D(y, p)\| + |H(x, p) - H(y, p)| \leq C_M(1 + |p|)|x - y|.$$

The Green-Kubo construction gives a correlation Hilbert factorization $D = \Sigma \Sigma^*$ with

$$\|\Sigma(x, p) - \Sigma(y, p)\|_{\text{HS}}^2 \leq C_M(1 + |p|)|x - y|.$$

Apply the Ishii matrix inequality to the vectors $(\Sigma(x, p)^*\xi, \Sigma(y, p)^*\xi)$ and pass through finite-rank projections in the correlation Hilbert bundle. This yields

$$\text{tr}(D(x, p)X) - \text{tr}(D(y, p)Y) \leq C_M\lambda|x - y|^2.$$

Combining the trace estimate with the Hamiltonian Lipschitz estimate gives the displayed modulus for $\mathcal{H}(y, p, Y) - \mathcal{H}(x, p, X)$. $\square$

Theorem 3.9 (Comparison for the derived HJB). *Let u be a bounded upper semicontinuous viscosity subsolution and let v be a bounded lower semicontinuous viscosity supersolution of*

$$(13) \quad \partial_t w + \mathcal{H}(x, Dw, D^2w) = 0$$

on $[0, T]$. If $u(T, \cdot) \leq v(T, \cdot)$ at the terminal time, then $u \leq v$. Consequently the effective HJB in Theorem 4.1 has at most one bounded viscosity solution in the comparison class used here.

Proof. Suppose that $\sup(u - v) = \delta > 0$. After the usual compact localization in the viscosity-comparison framework of [3, 5], choose $\lambda \rightarrow \infty$, $\beta \downarrow 0$, and $\eta \downarrow 0$, and maximize

$$u(t, x) - v(s, y) - \frac{\lambda}{2}|x - y|^2 - \frac{|t - s|^2}{2\beta} - \eta(T - t)^{-1}.$$

The terminal penalty and the terminal ordering force the doubled maximizer into the parabolic interior for small parameters. Standard doubled-variable estimates give

$$\lambda|x_\lambda - y_\lambda|^2 \rightarrow 0, \quad |t_\lambda - s_\lambda|^2/\beta \rightarrow 0.$$

The parabolic Ishii lemma supplies jets with common spatial gradient $p_\lambda = \lambda(x_\lambda - y_\lambda)$ and matrices X_λ, Y_λ satisfying the block inequality in Proposition 3.8. Applying the subsolution inequality to u , the supersolution inequality to v , and subtracting gives

$$\frac{\eta}{(T - t_\lambda)^2} \leq \omega_M(|x_\lambda - y_\lambda|(1 + |p_\lambda|) + \lambda|x_\lambda - y_\lambda|^2) + o_\beta(1).$$

The right-hand side tends to zero in the comparison limit, while the barrier term is strictly positive. This contradiction proves $u \leq v$. $\square$

4. DETERMINISTIC HJB LIMIT

Theorem 4.1 (Deterministic billiards to nonconvex HJB). *Assume the primitive billiard geometry, the finite-response mechanical port, and the companion theorem Paper 1, First-principles response package. Let u^ε be the prelimit viscosity-duality solution generated by the deterministic slow-fast billiard system. Then u^ε converges locally uniformly in the macroscopic variables, after the anisotropic microscopic pairing and Liouville-a.e. identification, to the unique viscosity solution u of*

$$(14) \quad \partial_t u + \mathcal{H}(x, Du, D^2u) = 0, \quad u(T, x) = \varphi(x).$$

Every coefficient in $\mathcal{H}$ is a deterministic transfer-resolvent output. No stochastic microscopic limit theorem, independence postulate, BSDE representation, or Girsanov formula is used in the derivation.

Proof. Define the upper and lower half-relaxed paired envelopes of u^ε on a fixed compact macroscopic cylinder and compact cotangent window. The prelimit viscosity-duality comparison estimate gives bounded macroscopic equicontinuity, while compactness of the admissible density bundle prevents escape of mass into grazing strips. The terminal condition passes directly because $u^\varepsilon(T, x, z) = \varphi(x)$ is independent of the microscopic state.

By Proposition 3.7, the upper envelope is a viscosity subsolution of the effective HJB and the lower envelope is a viscosity supersolution. The comparison theorem Theorem 3.9 gives upper envelope $\leq$ lower envelope; the opposite inequality is part of the definition of relaxed limits. Hence the two envelopes coincide with the unique viscosity solution u .

The residual constants in Lemma 3.5 are uniform over bounded admissible density balls and compact contact-jet windows, so the paired convergence is uniform on those windows. The Liouville-a.e. microscopic state statement follows by applying the paired convergence to a countable homogeneous-rectangle density basis and then using the deterministic branchwise oscillation modulus. The exceptional set is the union of the iterated singular orbit set with countably many density-basis null sets, and therefore has Liouville measure zero. No stochastic microscopic limit theorem enters the argument. $\square$

5. NONCONVEXITY AND THETA-EXPECTATIONS

Theorem 5.1 (Nonconvex theta-expectation). *There are finite-response billiard ports for which $p \mapsto H(x, p)$ is locally nonconvex. For such ports, the nonlinear semigroup generated by Theorem 4.1 is not generally subadditive:*

$$(15) \quad \mathcal{E}_t^\theta[\varphi + \psi] \neq \mathcal{E}_t^\theta[\varphi] + \mathcal{E}_t^\theta[\psi]$$

in general, and the subadditive inequality may fail in either direction.

Proof. Use the concrete port in Proposition 6.6 and choose x in the open set where $\mathbf{c}_e(x) = \frac{d^2}{ds^2}H(x, se)|_{s=0} < 0$. For small $a > 0$,

$$H(x, 2ae) - 2H(x, ae) = \mathbf{c}_e(x)a^2 + o(a^2) < 0.$$

Choose smooth terminal payoffs φ, ψ supported in a coordinate chart so that at x

$$D\varphi = D\psi = ae, \quad D^2\varphi = D^2\psi = 0.$$

For the terminal-value convention in Definition 5.2, the short-time viscosity expansion at such a smooth contact is

$$\mathcal{E}_t^\theta[\varphi](x) = \varphi(x) - tH(x, ae) + o(t), \quad \mathcal{E}_t^\theta[\varphi + \psi](x) = \varphi(x) + \psi(x) - tH(x, 2ae) + o(t).$$

Therefore

$$\mathcal{E}_t^\theta[\varphi + \psi](x) - \mathcal{E}_t^\theta[\varphi](x) - \mathcal{E}_t^\theta[\psi](x) = -t\{H(x, 2ae) - 2H(x, ae)\} + o(t) > 0$$

for small $t > 0$. This violates subadditivity. Changing the sign of the microscopic response direction gives the opposite Hamiltonian defect when that direction is desired, so the failure is not tied to a convex/sublinear special case. $\square$

Definition 5.2 (theta-expectation semigroup). For a terminal payoff φ , define

$$(16) \quad \mathcal{E}_t^\theta[\varphi](x) = u(0, x),$$

where u is the viscosity solution of the HJB equation in Theorem 4.1 on the interval $[0, t]$ with terminal payoff φ at time t .

Theorem 5.3 (Basic theta-expectation properties). *The semigroup $\mathcal{E}_t^\theta$ is constant preserving, monotone, and dynamically consistent. It is generally nonlinear and, by Theorem 5.1, it is not a subadditive G -expectation in the sense of [7] in general.*

6. CONCRETE EXAMPLE REQUIREMENT

Definition 6.1 (Concrete finite-horizon triangular Lorentz cell). Let Λ be the triangular lattice generated by

$$a_1 = (1, 0), \quad a_2 = (1/2, \sqrt{3}/2),$$

and let $\mathbb{T}_\Lambda^2 = \mathbb{R}^2/\Lambda$. Fix a circular scatterer centered at the lattice point with radius

$$r_0 \in \left(\frac{1}{2\sqrt{3}}, \frac{1}{2} \right).$$

The billiard cell is $\mathbb{T}_\Lambda^2 \setminus B(0, r_0)$. This is the standard finite-horizon triangular Lorentz cell: the lower bound on r_0 blocks the maximal triangular-lattice corridors, while the upper bound keeps neighboring scatterers disjoint. Small C^{r+3} normal deformations of the boundary stay inside the same finite-horizon dispersing class.

Definition 6.2 (Two-mode finite-response port on the concrete cell). Start from Definition 6.1. Write the circular boundary in arclength $s \in \mathbb{R}/(2\pi r_0 \mathbb{Z})$ and set

$$\beta(t) = \begin{cases} \exp(-1/(1-t^2)), & |t| < 1, \\ 0, & |t| \geq 1. \end{cases}$$

Choose $\delta < \pi r_0/12$ and define the normalized arclength bumps

$$\beta_{s_0, \delta}(s) = \frac{\beta(\text{dist}(s, s_0)/\delta)}{\|\beta(\text{dist}(\cdot, s_0)/\delta)\|_{C^{r+3}}}.$$

Set

$$\psi_1 = \beta_{0, \delta}, \quad \psi_2 = \beta_{2\pi r_0/3, \delta}.$$

These are disjoint nonnegative C^{r+3} normal deformation modes with C^{r+3} norm one. The deformed boundary is

$$q(s, \theta) = q(s, 0) + \theta_1 \psi_1(s) n(s) + \theta_2 \psi_2(s) n(s), \quad \theta \in [-\theta_*, \theta_*]^2.$$

For θ_* sufficiently small, the deformed cell remains strictly dispersing and finite horizon by openness of the curvature, separation, and corridor-blocking inequalities. For a fixed macroscopic direction $e \in \mathbb{R}^m$, define a microscopic finite-response selector

$$(17) \quad \Theta(x, p) = \theta_0 + \alpha_1(x)(p \cdot e)e_1 + \alpha_2(x)(p \cdot e)^2 e_2,$$

where e_1, e_2 are the coordinate directions in parameter space. The centered drift read-out and action read-out have branchwise smooth expansions

$$(18) \quad B(x, p, z) = B_0(x, z) + (p \cdot e)B_1(x, z) + (p \cdot e)^2 B_2(x, z) + O(|p|^3),$$

$$(19) \quad V(x, p, z) = V_0(x, z) - \lambda(x)\eta(z)(p \cdot e)^2 + \kappa(x)\eta(z)(p \cdot e)^4 + O(|p|^5),$$

with $\lambda(x) > 0$, $\kappa(x) > 0$, and a nonnegative branchwise smooth observable η with positive billiard average. These read-outs are part of the primitive finite-response port; they are fixed before the effective Hamiltonian is computed.

Definition 6.3 (Concrete observable and dominance constant). Let

$$\chi_\eta(s) = \beta_{4\pi r_0/3, \delta}(s).$$

This bump is supported on a boundary arc disjoint from the supports of ψ_1, ψ_2 . Define

$$(20) \quad \eta(s, \varphi) = \chi_\eta(s) \cos^2 \varphi.$$

Then $\langle \eta \rangle_{\theta_0} > 0$. Let $C_{\text{resp}}(x)$ be the Paper-1-exported response bound defined in Definition 6.4. It controls all second-order invariant-density and corrector-resolvent terms appearing in Proposition 6.6, excluding the direct action average. The dominance condition is

$$(21) \quad 2\lambda(x)\langle \eta \rangle_{\theta_0} > C_{\text{resp}}(x) + \delta_x \quad (\delta_x > 0).$$

Under this checkable inequality the directional Taylor coefficient is strictly negative.

Definition 6.4 (Paper 1 response bound used by the concrete example). For the concrete port, let the norms of the primitive read-outs be measured in the response domain exported by Paper 1 with the eight-derivative loss recorded in Paper 1, *Finite regularity-loss budget*. Denote by

$$\mathfrak{C}_{\Pi,2}, \quad \mathfrak{C}_{\mathcal{R},2}, \quad \mathfrak{C}_{\text{tr},2}, \quad \mathfrak{C}_{\text{loss}}$$

the uniform constants exported by the following Paper 1 components.

Constant	Paper 1 source
$\mathfrak{C}_{\Pi,2}$	Paper 1, <i>Smooth deterministic response</i> and Paper 1, <i>Uniform multi-response bounds</i> .
$\mathfrak{C}_{\mathcal{R},2}$	Paper 1, <i>Primitive spectral and resolvent theorem</i> , Paper 1, <i>Suspension Laplace contour and frequency partition</i> , and Paper 1, <i>Uniform multi-response bounds</i> .
$\mathfrak{C}_{\text{tr},2}$	Paper 1, <i>Trace insertion into resolvent words</i> together with the moving-boundary trace estimates.
$\mathfrak{C}_{\text{loss}}$	Paper 1, <i>Finite regularity-loss budget</i> .

They control, respectively, second derivatives of invariant projectors, second derivatives of reduced observable resolvents, moving-trace insertions in differentiated resolvent words, and the finite regularity-loss embedding. Define

$$(22) \quad C_{\text{resp}}(x) := \mathfrak{C}_{\text{loss}} \left[\mathfrak{C}_{\Pi,2} \|V_0(x, \cdot)\|_{r+8} + \mathfrak{C}_{\mathcal{R},2} (\|B_1(x, \cdot)\|_{r+8}^2 + \|B_2(x, \cdot)\|_{r+8}) \right. \\ \left. + \mathfrak{C}_{\text{tr},2} (\|\psi_1\|_{C^{r+8}} + \|\psi_2\|_{C^{r+8}}) (\|V_0(x, \cdot)\|_{r+8} + \|B_1(x, \cdot)\|_{r+8}) \right].$$

All constants in this display are downstream of Paper 1, *First-principles response package* and are produced before the HJB theorem. No effective Hamiltonian coefficient is included as input.

Proposition 6.5 (Response-bound provenance for the dominance test). *For the concrete port in Definition 6.2, the sum of all second-order invariant-density response terms and corrector/resolvent response terms in the directional coefficient $\mathfrak{c}_e(x)$ is bounded in absolute value by $C_{\text{resp}}(x)$.*

Proof. The second derivative of the invariant average is controlled by the projector derivative constant $\mathfrak{C}_{\Pi,2}$. The derivatives of the first corrector and of the second-order corrector are controlled by the differentiated reduced-resolvent words, hence by $\mathfrak{C}_{\mathcal{R},2}$. Terms in which the differentiated homogeneous partition or moving singularity boundary enters a resolvent word are controlled by Paper 1, *Trace insertion into resolvent words*, represented here by $\mathfrak{C}_{\text{tr},2}$. The downstream coefficient calculation consumes the derivative loss in Paper 1, *Finite regularity-loss budget*, represented here by $\mathfrak{C}_{\text{loss}}$. Multiplying these constants by the displayed primitive read-out norms gives exactly the bound in Definition 6.4. $\square$

Proposition 6.6 (Inspectable nonconvexity coefficient). *For the example target in Definition 6.2, define the deterministic Taylor coefficient*

$$(23) \quad \mathfrak{c}_e(x) = \left. \frac{d^2}{ds^2} H(x, se) \right|_{s=0},$$

where H is computed from the transfer-resolvent formula in Definition 3.2. The coefficient $\mathfrak{c}_e(x)$ is an explicit finite sum of:

- (i) direct action averages from $\partial_s^2 V(x, se, \cdot)|_{s=0}$;
- (ii) response derivatives of the invariant density under $\Theta(x, se)$;
- (iii) corrector/resolvent pairings involving $\partial_s B(x, se, \cdot)|_{s=0}$ and $\partial_s^2 B(x, se, \cdot)|_{s=0}$.

If the direct negative action term dominates the two response-corrector terms, so that $\mathfrak{c}_e(x) < 0$ on some open macroscopic set, then $p \mapsto H(x, p)$ is locally nonconvex there. Consequently Theorem 5.1

applies to concrete payoffs whose contact gradients lie in that set. For the concrete choice in Definition 6.3, the sufficient dominance condition is

$$2\lambda(x)\langle\eta\rangle_{\theta_0} > C_{\text{resp}}(x) + \delta_x.$$

Proof. Differentiate the deterministic coefficient formula in Definition 3.2 twice along $p = se$. Paper 1, *Smooth deterministic response* and Paper 1, *Moving-singularity response theorem* justify differentiating the invariant density, projectors, and reduced resolvent in the moving billiard fiber, while Paper 1, *Trace insertion into resolvent words* controls trace terms inside differentiated resolvent words. Thus the second derivative splits into the direct action term, the invariant-density response term, and the corrector/resolvent response term listed in the statement.

For the concrete read-out in Definition 6.2, the direct action contribution is

$$\left. \frac{d^2}{ds^2} \langle -\lambda(x)\eta s^2 \rangle_{\theta_0} \right|_{s=0} = -2\lambda(x)\langle\eta\rangle_{\theta_0}.$$

By Proposition 6.5, the absolute value of all remaining second-order response-corrector contributions is bounded by $C_{\text{resp}}(x)$. The dominance condition in Definition 6.3 therefore gives

$$\mathbf{c}_e(x) \leq -\delta_x < 0.$$

Taylor's formula then gives

$$H(x, se) + H(x, -se) - 2H(x, 0) = \mathbf{c}_e(x)s^2 + o(s^2) < 0$$

for all sufficiently small nonzero s . Hence $p \mapsto H(x, p)$ is locally nonconvex, and the same negative two-point defect is the short-time non-subadditivity input used in Theorem 5.1. $\square$

Citation-provenance status. The example is now inspectable at the level needed by Paper 2: the bump functions $\psi_1, \psi_2, \chi_\eta$ are explicit, the finite-horizon base table and dominance inequality are fixed in Definitions 6.1 and 6.3, and the analytic constant $C_{\text{resp}}(x)$ is tied to the named Paper 1 response theorems in Definition 6.4. The bibliography in this paper anchors only the macroscopic viscosity, homogenization, and nonlinear-expectation tools listed above. The singular-billiard literature is attached to the Paper 1 theorem statements rather than repeated here.

7. USE BY THE REPRESENTATION PAPER

The representation-calculus paper may take $\mathcal{E}_t^\theta$ and its generator as inputs. BSDE, Girsanov, and Cameron-Martin formulae should remain downstream representations and should not be used to prove Theorem 4.1.

APPENDIX A. TECHNICAL APPENDIX SCOPE

This appendix records the deterministic HJB proof details that are too long for the main article. It is not a second list of assumptions. The singular billiard estimates used below are exactly the response-theory outputs imported in the main paper: reduced observable resolvents, moving-boundary trace insertions, uniform differentiated resolvent words, suspension contour bounds, and finite regularity loss. The present appendix checks how those imported objects enter the finite-response action, the corrector hierarchy, the contact-frozen perturbed test, the half-relaxed viscosity passage, and the nonconvexity algebra.

Definition A.1 (HJB appendix dependency order). The proof details below use the following order.

- (i) The compact finite-response port fixes the table selector $\Theta(x, p)$, the centered drift $B(x, p, z)$, and the action read-out $V(x, p, z)$ before any effective coefficient is defined.
- (ii) Paper 1 supplies the invariant projector, reduced flow resolvent, and moving-response bounds for the selected billiard fiber.

- (iii) The first and second cell equations are solved in the zero-mean observable space of that fiber.
- (iv) The prelimit Hamilton-Jacobi operator is tested only after the actual viscosity contact jet has been frozen.
- (v) The effective HJB equation is identified by the invariant part of the order-one residual; the centered part is cancelled by the second corrector.
- (vi) The upper and lower relaxed limits are compared by the macroscopic viscosity comparison theorem.

Thus the HJB equation is an output of the deterministic calculation, not an input to the microscopic port or to the response theorem.

APPENDIX B. FINITE-RESPONSE MICRO-ACTION

Definition B.1 (Microscopic finite-response port). The primitive finite-response device consists of a compact internal energy surface $\mathcal{Y}$, a deterministic finite-time response flow

$$\mathcal{R}_s^{x,P} : \mathcal{Y} \rightarrow \mathcal{Y}, \quad 0 \leq s \leq s_{\text{fr}},$$

a smooth saturation $\Pi_{\text{fr}} : \mathbb{R}^m \rightarrow \mathbb{R}^m$, and branchwise smooth read-out maps

$$\vartheta(x, y), \quad \beta(x, y, z), \quad \mathcal{V}(x, y, z).$$

With y_0 the calibrated rest state and $y(x, P) = \mathcal{R}_{s_{\text{fr}}}^{x,P} y_0$, set

$$\Theta(x, p) = \vartheta(x, y(x, \Pi_{\text{fr}}(p))), \quad \widehat{B}(x, p, z) = \beta(x, y(x, \Pi_{\text{fr}}(p)), z), \quad V(x, p, z) = \mathcal{V}(x, y(x, \Pi_{\text{fr}}(p)), z).$$

The centered drift used in the moving slow frame is

$$B(x, p, z) = \widehat{B}(x, p, z) - \int \widehat{B}(x, p, \zeta) d\mu_{\Theta(x,p)}(\zeta).$$

All derivatives needed below are bounded on compact contact-jet windows and in homogeneous flow boxes. On the physical cotangent window $\Pi_{\text{fr}}(p) = p$, so local Taylor coefficients in p are genuine mechanical response coefficients.

Proposition B.2 (No effective-coefficient feedback). *The data Θ, B, V in Definition B.1 are fixed before the correctors, Green-Kubo tensor, Hamiltonian, HJB solution, and theta-expectation semigroup are introduced. The deterministic construction factors as*

$$(\mathcal{Y}, \mathcal{R}, \Pi_{\text{fr}}, \vartheta, \beta, \mathcal{V}) \longmapsto (\Theta, B, V) \longmapsto (\Pi_{\theta}, \mathcal{R}_{\theta}, \chi, \psi) \longmapsto (D, H),$$

where $\mathcal{R}_{\theta}$ denotes the reduced observable resolvent supplied by Paper 1. No arrow points from D, H , from a viscosity solution u , or from $\mathcal{E}^{\theta}$ back to the finite-response port.

Proof. The first arrow is a finite-dimensional deterministic operation: solve the internal response flow for the finite time s_{fr} , read the table parameter, force, and action density, and subtract the instantaneous microcanonical velocity. The second arrow is the companion response theorem: for the table $\Theta(x, p)$, it constructs the invariant projector and the zero-mean reduced resolvent. The third arrow solves the cell equations and takes invariant pairings. Since the HJB operator appears only in the last arrow, it cannot be used to define any earlier object. $\square$

Lemma B.3 (Exact endpoint jets from a compact port). *Fix a compact physical cotangent window K_{phys} and a finite order $J \leq r - 6$. Let $\mathcal{J}(P)$ be any finite list of saturated polynomial jets in P satisfying the microscopic parity, zero-gauge, and smallness constraints of the finite-response port. Then the compact internal response system can be enlarged by finitely many coordinates so that its endpoint map*

$$E_A(x, P) = y(s_{\text{fr}}; x, P, A)$$

has exactly the prescribed J -jet on K_{phys} . The construction is triangular in the jet degree and is completed before invariant densities, cell equations, or effective coefficients are introduced.

Proof. Choose saturated monomials $M_\alpha(P) = \Pi_{\text{fr}}(P)^\alpha$, which agree with ordinary monomials on K_{phys} . In a compact coordinate ball of the internal energy surface, add linear forcing coordinates with amplitudes $A_{\alpha\ell}(x)M_\alpha(P)$. Picard iteration of the deterministic finite-time ODE gives

$$E_A(x, P) = \frac{s_{\text{fr}}^2}{2} \sum_{\alpha, \ell} A_{\alpha\ell}(x) M_\alpha(P) e_\ell + O(|A|^2).$$

At degree q the lower-degree amplitudes have already determined a finite polynomial error in the endpoint jet. Adding a fresh internal coordinate with linear force equal to the negative of that error changes no lower-degree jet and cancels the degree- q error. The finite Jacobian from the new amplitudes to the degree- q endpoint coefficients is diagonal with nonzero entries $s_{\text{fr}}^2/2$, so the recursion terminates.

The local potential is then cut off inside the internal compact energy surface and saturated outside K_{phys} . Smallness of the endpoint amplitudes keeps the scatterer deformation in the geometric window. The construction is therefore a finite-dimensional endpoint controllability statement; it does not use the transfer spectrum, the Green-Kubo tensor, the Hamiltonian, or a limiting HJB solution. $\square$

Proposition B.4 (Endpoint reciprocity ledger). *The table displacement Θ , centered drift B , and action read-out V are generated by one finite-response action. On each homogeneous branch there are zero-mean branch gauges $R_{\text{fr}}^0, R_{\text{fr}}^a$ such that*

$$B^a(x, p, z) = -\partial_{x_a} \mathcal{S}_{\text{fr}}(x, \Pi_{\text{fr}}(p); z) + \mathcal{A}_{\Theta(x, p)} R_{\text{fr}}^a(x, p, z),$$

$$V(x, p, z) = \partial_{s_{\text{fr}}} \mathcal{S}_{\text{fr}}(x, \Pi_{\text{fr}}(p); z) + \mathcal{A}_{\Theta(x, p)} R_{\text{fr}}^0(x, p, z),$$

and

$$D_p \Theta(x, p) = D_y \vartheta(x, E(x, \Pi_{\text{fr}}(p))) D_P E(x, \Pi_{\text{fr}}(p)) D \Pi_{\text{fr}}(p).$$

Consequently the finite Taylor coefficients used by the concrete port obey ordinary action reciprocity identities; they are not independent effective HJB coefficients.

Proof. Hamilton's endpoint variation formula for the finite-response action gives the slow force from x -variation, the running action from response-time variation, and the table displacement from endpoint variation. Billiard flight endpoint changes add only generator coboundaries, represented by the gauge terms above; their invariant means vanish because $\int \mathcal{A}_\theta R d\mu_\theta = 0$ on the generator domain. Differentiating the endpoint formulas gives the stated reciprocity relations. Since Lemma B.3 realizes the endpoint jet before any cell-resolvent object is formed, these relations are microscopic constraints rather than post-hoc choices of D or H . $\square$

Definition B.5 (Prelimit viscosity-duality operator). On a fixed contact-jet window, the branch-wise prelimit operator has the fast part

$$\mathcal{L}_{\text{fast}}^\varepsilon \Phi = -\varepsilon^{-2} \mathcal{A}_{\Theta(x, p)} \Phi - \varepsilon^{-1} B(x, p, z) \cdot D_x \Phi,$$

and an order-one action part

$$\mathcal{L}_{\text{act}}^\varepsilon \Phi = -V^\circ(x, p, z) + \text{finite-response work terms generated by the full slow gradient.}$$

Here p is the physical cotangent signal read by the port. In the viscosity argument p is frozen at the actual contact gradient before the singular fast terms are cancelled. The operator is interpreted in anisotropic duality against admissible densities in the billiard fiber.

APPENDIX C. EXACT PRELIMIT ACTION GRAPH

Definition C.1 (Branchwise action graph). Fix $\varepsilon > 0$, a terminal payoff ϕ , and a regular homogeneous flow box in the universal billiard bundle. A prelimit action graph is a branchwise C^1 graph

$$\Lambda_s^\varepsilon = \{(x_s, z_s, P_s, A_s) : P_s = D_x A_s, (x_s, z_s) \in U_{\text{br}}\}$$

whose dynamics on the branch are

$$\dot{z}_s = \varepsilon^{-2} F_{\Theta(x_s, P_s)}(z_s), \quad \dot{x}_s = \varepsilon^{-1} B(x_s, P_s, z_s), \quad \dot{A}_s = V^\circ(x_s, P_s, z_s).$$

The signal P_s is the cotangent coordinate of this same graph. It is not chosen from the later macroscopic solution.

Proposition C.2 (Closed prelimit action graph). *For every compact contact-jet window and every terminal graph $A_T(x, z) = \phi(x)$, $P_T = D\phi(x)$, the branchwise action graph in Definition C.1 exists and is unique up to the first exit time from a homogeneous flow box. On finite time intervals the graph is obtained by concatenating finitely many such branch solutions and by ignoring only the canonical zero-flux singular set.*

Proof. On a homogeneous flow box the finite-response read-outs and the pulled-back billiard vector field are C^{r-4} in (x, P, z) and uniformly Lipschitz on the compact window. The branch ODE therefore has a unique solution. The cotangent relation is preserved because the same deterministic action generates the slow force, the finite-response table displacement, and the running cost: differentiating the action identity with respect to the initial slow coordinate gives the adjoint equation for P_s . Specular reflection preserves the canonical one-form, since endpoint variations are tangent to the boundary while the reflected normal velocity changes sign. Grazing and multiple-collision hypersurfaces are not places where the pointwise equation is evaluated; they have zero flux measure and later enter only through weak trace estimates. $\square$

Theorem C.3 (Exact branchwise prelimit Hamilton-Jacobi identity). *On every regular branch, the principal function generated by Proposition C.2 satisfies*

$$-\partial_t U^\varepsilon - \varepsilon^{-2} \mathcal{A}_{\Theta(x, p)} U^\varepsilon - \varepsilon^{-1} B(x, p, z) \cdot p - V^\circ(x, p, z) = 0, \quad p = D_x U^\varepsilon(t, x, z).$$

After completion in the observable-duality domain this is the prelimit viscosity-duality equation used in the main paper. It is constructed before the cell problem, the Green-Kubo tensor, the effective Hamiltonian, or the theta-expectation semigroup.

Proof. For a small branch time $h > 0$, the action identity gives

$$U^\varepsilon(t, x, z) = U^\varepsilon(t + h, x_h, z_h) + \int_t^{t+h} V^\circ(x_s, P_s, z_s) ds + o(h).$$

Taylor expansion of the right-hand side along the branch trajectory produces the time derivative, the fast generator term, and the slow work term $\varepsilon^{-1} B \cdot D_x U^\varepsilon$. Dividing by h and sending $h \downarrow 0$ gives the displayed identity. The anisotropic completion is obtained by applying this branch identity against admissible densities and closing the Green identity with the same moving-boundary trace estimates imported from Paper 1. $\square$

Lemma C.4 (Universal-bundle meaning of a nonconstant contact covector). *If a perturbed test has contact covector*

$$P_\Phi(t, x, z) = D_x \Phi(t, x, z),$$

then the prelimit operator evaluated at P_Φ is a single branch vector field on the universal billiard bundle. It is not a simultaneous evolution of many unrelated billiard tables. The map

$$(x, P, z) \mapsto \mathcal{A}_{\Theta(x, P)} + \varepsilon^{-1} B(x, P, z) \cdot P + V^\circ(x, P, z)$$

is evaluated along the graph $z \mapsto P_\Phi(t, x, z)$, and its derivative in that graph direction is a strong-to-weak moving-response term.

Proof. Normal-coordinate transport identifies nearby tables with one reference bundle. In that bundle, the impact equation, reflection law, roof, and Jacobian are branchwise functions of (x, P, z) . Substituting a branchwise field $P_\Phi(z)$ is therefore ordinary composition before the singular sets are completed. Differentiating the composed vector field moves the branch maps and the singular hypersurfaces; these contributions are exactly the interior, roof, and trace letters controlled by the companion response theorem. $\square$

Lemma C.5 (Specular endpoint variation). *On a regular homogeneous branch, differentiating the Hamilton principal action through a collision creates no extra reflection work. If q is the collision point, v^- and v^+ are the incoming and outgoing unit velocities, and $\delta q \in T_q \partial Q_{\Theta(x,p)}$, then*

$$\langle v^- - v^+, \delta q \rangle = 0.$$

Thus a regular specular reflection preserves the canonical endpoint term in the branch action.

Proof. Specular reflection preserves the tangential velocity component and reverses only the normal component:

$$v^+ = v^- - 2\langle v^-, n(q) \rangle n(q).$$

The jump $v^- - v^+$ is therefore normal to the boundary, while a regular endpoint variation δq is tangent. Their pairing vanishes. The Liouville flux on the two physical traces is the same with opposite normal orientation, so the Green identity has no physical-boundary residue. $\square$

Proposition C.6 (Moving branch-boundary ledger). *The only boundary terms left after Lemma C.5 are derivatives of branch selectors: grazing faces, homogeneity boundaries, and branch-change hypersurfaces. In the viscosity-duality pairing these terms are weak trace letters. On a compact contact-jet window,*

$$\sum_{\Gamma \subset \mathcal{S}_{\text{sing}}} | \langle [F]_\Gamma, (X_{x,P} \cdot n_\Gamma) \rho \rangle_\Gamma | \leq C_{\text{tr}}(M, K) \|F\|_\# \|\rho\|_{\mathcal{A}_M^{\text{adm}}} + C_{\text{tail}}(M, K, N),$$

where $C_{\text{tail}}(M, K, N) \rightarrow 0$ as the homogeneity cutoff $N \rightarrow \infty$.

Proof. After normal-coordinate trivialization, a moving homogeneous branch is selected by a characteristic function. Differentiating it gives a hypersurface distribution supported on the moved singular curve. The normal speed of this hypersurface is controlled by the same impact equation and crossing denominator used in the response theorem. Central strips give the trace constant C_{tr} . In strip m , the denominator growth is polynomial in m , while the admissible density tail and the response strip weight are summable. This gives the cutoff tail C_{tail} . Hence moving boundaries enter only through controlled weak traces, not through unaccounted endpoint work. $\square$

APPENDIX D. CORRECTOR HIERARCHY

Definition D.1 (First corrector and Green-Kubo tensor). Fix (x, p) and write $\theta = \Theta(x, p)$. For the centered drift components $B_i(x, p, \cdot)$, define

$$\chi_i(x, p, \cdot) = -\mathcal{A}_\theta^{-1} B_i(x, p, \cdot), \quad \Pi_\theta \chi_i = 0.$$

For $a \in \mathbb{R}^m$, the quadratic Green-Kubo form is

$$a^T D(x, p) a = \int_0^\infty \int (a \cdot B)(z) (a \cdot B)(\Phi_\theta^t z) d\mu_\theta(z) dt,$$

equivalently

$$D_{ij}(x, p) = \frac{1}{2} \int (B_i \chi_j + B_j \chi_i) d\mu_\theta$$

after the finite-response work connection is reduced by the single-action reciprocity identity.

Lemma D.2 (First-order cancellation at the actual contact jet). *Let $\varphi \in C^3$ and let $(t_\varepsilon, x_\varepsilon)$ be the prelimit contact point. Set*

$$p_\varepsilon = D\varphi(t_\varepsilon, x_\varepsilon), \quad \theta_\varepsilon = \Theta(x_\varepsilon, p_\varepsilon),$$

and

$$\chi_{p_\varepsilon} = \sum_i p_{\varepsilon,i} \chi_i(x_\varepsilon, p_\varepsilon, \cdot).$$

Then the scale- ε^{-1} part of $\mathcal{L}^\varepsilon(\varphi + \varepsilon \chi_{p_\varepsilon})$ vanishes:

$$-\varepsilon^{-2} \mathcal{A}_{\theta_\varepsilon}(\varepsilon \chi_{p_\varepsilon}) - \varepsilon^{-1} B(x_\varepsilon, p_\varepsilon, z) \cdot p_\varepsilon = 0.$$

No term of the form $\varepsilon^{-1}(p_\varepsilon - p_0)$ is produced, because the cell equation is solved after the actual prelimit contact jet is known.

Proof. The first corrector solves

$$\mathcal{A}_{\theta_\varepsilon} \chi_{p_\varepsilon} = -B(x_\varepsilon, p_\varepsilon, \cdot) \cdot p_\varepsilon.$$

Substitution gives the displayed cancellation in the observable dual space. The limiting jet (x_0, p_0) is not inserted until after this cancellation has been made, so there is no singular mismatch. $\square$

Definition D.3 (Contact-frozen perturbed test). At the same contact point define

$$X_\varepsilon = D^2\varphi(t_\varepsilon, x_\varepsilon), \quad \chi_\varepsilon = (\chi_1, \dots, \chi_m)^T(x_\varepsilon, p_\varepsilon, \cdot).$$

The first corrector is frozen in its coefficient arguments but multiplied by the current macroscopic gradient:

$$\Xi_\varepsilon[\varphi](t, x, z) = \sum_i \partial_i \varphi(t, x) \chi_i(x, p_\varepsilon, z).$$

At the contact point its full slow gradient is

$$q_\varepsilon(z) = D_x^{\text{fr}} \chi_{p_\varepsilon}(x_\varepsilon, p_\varepsilon, z) + X_\varepsilon^T \chi_\varepsilon(z).$$

The second term is the Hessian contribution from the complete chain rule; it is not discarded and it is not replaced by a pointwise smallness claim.

Definition D.4 (Complete order-one work observable). Let $\mathfrak{W}_{x,p}(q)$ denote the finite-response work observable created by inserting a full slow-gradient increment $q(z)$ in the port:

$$\mathfrak{W}_{x,p}(q) = B(x, p, \cdot) \cdot q + (\nabla_\theta \mathcal{A}_\theta)[D_p \Theta(x, p) q] \chi_p + (\nabla_p B(x, p, \cdot)[q]) \cdot p.$$

Its invariant and centered parts are

$$\Lambda_{x,p}(q) = \int \mathfrak{W}_{x,p}(q) d\mu_{\Theta(x,p)}, \quad \mathfrak{W}_{x,p}^\circ(q) = \mathfrak{W}_{x,p}(q) - \Lambda_{x,p}(q).$$

For the contact-frozen test, the order-one observable is

$$\mathcal{G}_\varepsilon^\varphi(z) = V^\circ(x_\varepsilon, p_\varepsilon, z) + \mathfrak{W}_{x_\varepsilon, p_\varepsilon}(q_\varepsilon)(z).$$

Proposition D.5 (Order-one average and second cell equation). *The invariant part of $\mathcal{G}_\varepsilon^\varphi$ is exactly*

$$\int \mathcal{G}_\varepsilon^\varphi d\mu_{\theta_\varepsilon} = \text{Tr}(D(x_\varepsilon, p_\varepsilon) X_\varepsilon) + H(x_\varepsilon, p_\varepsilon).$$

The centered part is solved by the deterministic second corrector

$$\psi_\varepsilon = -\mathcal{A}_{\theta_\varepsilon}^{-1} \Pi_{\theta_\varepsilon}^\perp \mathcal{G}_\varepsilon^\varphi, \quad \Pi_{\theta_\varepsilon} \psi_\varepsilon = 0.$$

Thus the order-one microscopic oscillation is cancelled before the half-relaxed limit is taken.

Proof. The horizontal component of q_ε contributes the finite-response work term in H . The Hessian component $X_\varepsilon^T \chi_\varepsilon$ contributes

$$\sum_{i,j} X_{\varepsilon,ij} \Lambda_{x_\varepsilon, p_\varepsilon}(\chi_i e_j).$$

The single-action reciprocity of the finite-response port makes the connection part skew in i, j . Since X_ε is symmetric, the skew part vanishes after contraction and the remaining symmetric part is the Green-Kubo tensor. The residual centered part has zero invariant projection and is in the range of the reduced observable generator by the Paper 1 resolvent theorem. $\square$

Definition D.6 (Full corrector hierarchy window). For a contact jet (x, p, X) , define the hierarchy by the recursive rule

$$\mathcal{A}_{\Theta(x,p)} \chi_a^{(1)} = -\Pi_{\Theta(x,p)}^\perp B_a(x, p, \cdot), \quad \Pi_{\Theta(x,p)} \chi_a^{(1)} = 0,$$

and, after all order- k invariant averages have been separated,

$$\mathcal{A}_{\Theta(x,p)} \chi^{(k+1)} = -\Pi_{\Theta(x,p)}^\perp \mathcal{G}^{(k)}(x, p, X, \chi^{(1)}, \dots, \chi^{(k)}), \quad \Pi_{\Theta(x,p)} \chi^{(k+1)} = 0.$$

For the HJB limit in the main paper, only $k = 1$ and $k = 2$ are needed. The recursive notation records that every higher remainder used for Taylor bookkeeping would be obtained by the same reduced resolvent on the zero-mean observable space.

Proposition D.7 (Hierarchy closure and branch summation). *On compact contact-jet windows, every corrector and every differentiated corrector appearing in Definition D.6 is a finite sum of reduced-resolvent words applied to branchwise finite-response observables and moving-trace letters. For each fixed order $k \leq r - 8$,*

$$\|\chi^{(k)}\|_\# + \sum_{|\alpha| \leq r-8-k} \|\partial_{x,p}^\alpha \chi^{(k)}\|_w \leq C_k(K).$$

The constants are obtained branchwise and then summed over homogeneity children by the same growth/strip-tail mechanism as in Lemma F.2.

Proof. The datum $\mathcal{G}^{(k)}$ is built from products of finite-response read-outs, previous correctors, and horizontal derivatives. Product bounds in the anisotropic observable space put its centered part in the domain of the reduced generator. Applying the reduced resolvent gives $\chi^{(k+1)}$. When a parameter derivative hits the projector, generator, roof, or singular branch selector, the resolvent identity gives an ordered word with one differentiated letter. Interior letters are branchwise smooth; boundary letters are exactly the trace functionals of Proposition C.6. The high-strip polynomial loss is fixed at each finite order, and the admissible strip tail makes the global child sum finite. This proves the displayed bounds by induction on k . $\square$

APPENDIX E. RESIDUAL IDENTITY AND SCALE LEDGER

Lemma E.1 (Residual identity with normalized gauge). *Let*

$$\Phi^\varepsilon = \varphi + \varepsilon \Xi_\varepsilon[\varphi] + \varepsilon^2 \psi_\varepsilon$$

and normalize it by subtracting the microscopic gauge value at the contact point. When Φ^ε is inserted into the prelimit viscosity-duality operator and paired with any admissible density $\rho \in \mathcal{A}_M^{\text{adm}}$, the result is

$$\partial_t \varphi(t_\varepsilon, x_\varepsilon) + \mathcal{H}(x_\varepsilon, p_\varepsilon, X_\varepsilon) + \langle R_\varepsilon[\varphi], \rho \rangle,$$

where

$$\mathcal{H}(x, p, X) = -\text{Tr}(D(x, p)X) - H(x, p)$$

and, on compact contact-jet windows,

$$|\langle R_\varepsilon[\varphi], \rho \rangle| \leq C_M(\varepsilon + |t_\varepsilon - t_0| + |x_\varepsilon - x_0| + |p_\varepsilon - p_0| + |X_\varepsilon - X_0|).$$

Proof. The scale- ε^{-1} terms cancel by Lemma D.2. The invariant order-one part is identified by Proposition D.5, and its centered part is cancelled by the second cell equation. Every remaining term has one of the following forms:

Source	Analytic form	Bound
Smooth test expansion	third-order Taylor and contact displacement terms	C times contact displacement
First corrector remainder	$\varepsilon \partial_t \Xi_\varepsilon$ and horizontal products	$C_M \varepsilon$
Second corrector remainder	$\varepsilon B \cdot D_x \psi_\varepsilon$ and time derivatives	$C_M \varepsilon$
Moving singular boundaries	trace distributions from changing branch charts	$C_M \varepsilon$ after strip-tail summation
Parameter continuity	replacement of the actual contact jet by the limiting jet after cancellation	C times contact displacement

Product, trace, and differentiated-resolvent bounds are exactly the Paper 1 inputs. Summing the displayed finite list proves the estimate. $\square$

Proposition E.2 (Four-scale deterministic ledger). *The perturbed-test calculation is accounted for by*

$$\varepsilon^{-2} : \mathcal{A}_\theta \varphi = 0 \quad (\varphi \text{ has no fast variable}),$$

$$\varepsilon^{-1} : \mathcal{A}_\theta \chi_p + B \cdot p = 0,$$

$$1 : V^\circ + \mathfrak{W}(q) \text{ has mean } \text{Tr}(DX) + H \text{ and centered part solved by } \psi,$$

$$\varepsilon : \text{Taylor, horizontal, roof, and moving-trace remainders.}$$

The order-one line uses the actual complete gradient increment

$$q = D_x^{\text{fr}} \chi_p + X^T \chi.$$

Therefore neither a hidden ε^{-1} parameter mismatch nor an omitted $D_p \chi X$ chain-rule contribution remains.

Proof. The ledger is the combination of Lemmas D.2 and E.1, Definition D.3, and Proposition D.5. The crucial point is the order of operations: freeze the actual contact jet, cancel the singular powers, split the order-one term into invariant and centered parts, and only then pass to the limiting jet. $\square$

APPENDIX F. BRANCH RESIDUAL CONSTANTS

Definition F.1 (Residual constant ledger). Fix an admissible density radius M and a compact contact-jet window K . The residual estimate in Lemma E.1 uses the following constants:

Constant	Role
$C_{\text{fr}}(K)$	bounds finite-response derivatives of Θ, B, V on K .
$C_{\text{res}}(K)$	bounds the reduced flow resolvent and the first and second cell correctors.
$C_{\text{word}}(K)$	bounds differentiated resolvent words generated by parameter and contact-jet derivatives.
$C_{\text{prod}}(M, K)$	bounds products of admissible densities with branchwise correctors and horizontal derivatives.
$C_{\text{tr}}(M, K)$	bounds moving-singularity trace letters in the weak anisotropic norm.
$C_{\text{tail}}(M, K, N)$	bounds the high-strip tail after strip cutoff N , with $C_{\text{tail}}(M, K, N) \rightarrow 0$ as $N \rightarrow \infty$.
$C_{\text{ct}}(K)$	bounds the contact displacement modulus for (t, x, p, X) in K .

Choose $N = N(M, K, \eta)$ so that $C_{\text{tail}}(M, K, N) \leq \eta$, and then choose ε small enough that all central-strip, Taylor, and corrector remainders are below η . The resulting residual modulus is

$$\Omega_{M,K}(\varepsilon, \Delta_{\text{ct}}) = C_0(M, K)\varepsilon + C_{\text{ct}}(K)\Delta_{\text{ct}} + C_{\text{tail}}(M, K, N(\varepsilon)),$$

where

$$\Delta_{\text{ct}} = |t_\varepsilon - t_0| + |x_\varepsilon - x_0| + |p_\varepsilon - p_0| + |X_\varepsilon - X_0|.$$

For the contact points selected in the half-relaxed argument, $\Omega_{M,K}(\varepsilon, \Delta_{\text{ct}}) \rightarrow 0$.

Lemma F.2 (Branch residual summation). *Every residual term in the normalized perturbed-test calculation is a finite sum of homogeneous-branch contributions bounded by the constants in Definition F.1. After summing over branch children,*

$$|\langle R_\varepsilon[\varphi], \rho \rangle| \leq \Omega_{M,K}(\varepsilon, \Delta_{\text{ct}})(1 + \|\varphi\|_{C^4})$$

for all $\rho \in \mathcal{A}_M^{\text{adm}}$.

Proof. On a central homogeneous branch, the Taylor remainder and the finite-response coefficient remainder are ordinary C^4 estimates with constants C_{fr} and C_{ct} . The first and second corrector remainders are products of admissible densities, correctors, and horizontal derivatives; they are bounded by $C_{\text{prod}}C_{\text{res}}\varepsilon$. Differentiating the parameter-dependent projector or resolvent gives finitely many ordered resolvent words, bounded by C_{word} .

When the branch cut moves, the residual is a weak trace functional supported on a singular hypersurface. The trace bound C_{tr} controls its central strip contribution. In high strips, the crossing denominator grows only as a fixed polynomial in the strip index, while admissible densities carry the strip-tail weight supplied by the response theorem. Hence the total high-strip contribution is $C_{\text{tail}}(M, K, N)$. Summing first over central branches and then over the high-strip tail gives the displayed bound. $\square$

Definition F.3 (Branch residual family decomposition). For a fixed admissible-density radius M , compact contact-jet window K , and strip cutoff N , decompose the normalized residual as

$$R_\varepsilon = R_\varepsilon^{\text{cent}} + R_\varepsilon^{\text{word}} + R_\varepsilon^{\text{tr}} + R_\varepsilon^{\text{end}} + R_\varepsilon^{\text{tail}}.$$

The five terms are defined by their source after freezing the contact jet:

Term	Source
$R_\varepsilon^{\text{cent}}$	central-branch Taylor, roof, and smooth finite-response coefficient remainders.
$R_\varepsilon^{\text{word}}$	differentiated projector, generator, reduced-resolvent, and corrector words with all letters supported in the central strip family.
$R_\varepsilon^{\text{tr}}$	moving grazing, homogeneity-boundary, and branch-change trace letters in central strips.
$R_\varepsilon^{\text{end}}$	endpoint and terminal-gauge terms, including specular endpoint variations and cash-additive terminal shifts.
$R_\varepsilon^{\text{tail}}$	all central estimates repeated in strips above N , plus their terminal children.

This decomposition is made after the scale- ε^{-1} cancellation and after the order-one invariant average has been separated. Hence no term in the table changes the effective coefficients; the table only audits the vanishing residual.

Lemma F.4 (Central branch residual check). *For each central homogeneous branch b below the cutoff N ,*

$$|\langle R_{\varepsilon,b}^{\text{cent}} + R_{\varepsilon,b}^{\text{word}} + R_{\varepsilon,b}^{\text{end}}, \rho \rangle| \leq C_{M,K,N}(1 + \|\varphi\|_{C^5})(\varepsilon + \Delta_{\text{ct}})$$

uniformly for $\rho \in \mathcal{A}_M^{\text{adm}}$. The constant depends on N , but not on ε , the branch label b , or the selected upper/lower contact.

Proof. On a central branch the collision map, free-flight roof, finite-response selector, and branch Jacobian are C^{r-6} functions of the contact jet. The Taylor remainder of the branch principal action is therefore bounded by $C_{K,N}\|\varphi\|_{C^5}(\varepsilon + \Delta_{\text{ct}})$. If a derivative hits the invariant projector, reduced generator, or corrector, the resolvent identity produces a finite ordered word. Every letter is central

and has bounded strong or weak norm on K , so the product bound gives the same factor. Endpoint terms are handled by Lemma C.5 and Proposition I.9: the physical reflection contributes no extra boundary work, while a terminal constant changes only the cash-additive level and not the residual derivatives. There are finitely many central branch families below N , which gives the displayed uniform constant. $\square$

Lemma F.5 (Trace branch residual check). *The central-strip moving-boundary part satisfies*

$$|\langle R_\varepsilon^{\text{tr}}, \rho \rangle| \leq C_{\text{tr}}(M, K, N)(1 + \|\varphi\|_{C^5})(\varepsilon + \Delta_{\text{ct}})$$

for every admissible density $\rho \in \mathcal{A}_M^{\text{adm}}$. The trace constant is the same for upper and lower half-relaxed contacts.

Proof. Moving singular hypersurfaces enter only when differentiating a branch selector or a strip cutoff. In local normal coordinates the derivative is a coarea trace with a normal crossing denominator. The finite-response port keeps the crossing speed and the contact jet in a compact window, so the central denominator loss is bounded by $C_{\text{tr}}(M, K, N)$. The trace is a weak anisotropic functional, exactly the class controlled by the imported moving-response theorem. Upper and lower contacts use the same branch selectors and the same normalized gauge; only the sign of the final viscosity inequality changes, not the trace norm. $\square$

Proposition F.6 (High-strip residual cutoff). *For every $\eta > 0$ there is a cutoff $N = N(M, K, \eta)$ such that*

$$|\langle R_\varepsilon^{\text{tail}}, \rho \rangle| \leq \eta(1 + \|\varphi\|_{C^5})$$

for all sufficiently small ε , uniformly over all contacts in K and all $\rho \in \mathcal{A}_M^{\text{adm}}$.

Proof. In strip n , differentiating the branch action can lose only the fixed polynomial power already listed in the Paper 1 response budget: impact time, roof, branch Jacobian, trace multiplicity, and the finite number of corrector word letters. The admissible density class carries a strip-tail weight with a larger exponent. Summing over terminal descendants first gives a geometric standard-pair factor, while the remaining polynomial strip sum tends to zero as $N \rightarrow \infty$. The choice of N is made before sending $\varepsilon \rightarrow 0$, so the cutoff is uniform for the half-relaxed passage. $\square$

Proposition F.7 (Full branch residual check). *With the decomposition of Definition F.3, the residual modulus in Lemma F.2 may be chosen as*

$$\Omega_{M,K}(\varepsilon, \Delta_{\text{ct}}) = C_{M,K,N}(1 + \|\varphi\|_{C^5})(\varepsilon + \Delta_{\text{ct}}) + \eta(1 + \|\varphi\|_{C^5}),$$

where $N = N(M, K, \eta)$. Consequently, after first choosing N and then letting $\varepsilon \rightarrow 0$, the branch residual vanishes uniformly in the subsolution and supersolution passages.

Proof. Add the five terms in Definition F.3. Central smooth, word, and endpoint terms are controlled by Lemma F.4. Central moving-boundary traces are controlled by Lemma F.5. The high-strip remainder is controlled by Proposition F.6. These bounds are uniform over the paired upper/lower contacts used in Lemma H.1; therefore the order of limits gives the displayed modulus and the vanishing residual required by Proposition H.2. $\square$

Proposition F.8 (Uniform sub/sup residual modulus). *The subsolution and supersolution inequalities in Proposition H.2 hold with the same residual modulus $\Omega_{M,K}$. The sign of the viscosity inequality changes between the two halves, but the constants and the branch summations are identical.*

Proof. Both upper and lower contacts use the same normalized gauge, the same contact-frozen first corrector, and the same second cell equation. The branchwise prelimit operator is the same deterministic action operator in both cases. Therefore the residual sources listed in Definition F.1 are unchanged. Only the inequality direction in the viscosity-duality test changes, so the same $\Omega_{M,K}$ controls both limits. $\square$

APPENDIX G. SMOOTH APPROXIMATION IN THE WEAK TOPOLOGY

Lemma G.1 (Admissible density smoothing). *Fix an admissible-density ball $\mathcal{A}_M^{\text{adm}}$, a compact contact-jet window K , and $\eta > 0$. For every $\rho \in \mathcal{A}_M^{\text{adm}}$ there is a density ρ_η that is C^{r-6} on each homogeneous rectangle, vanishes in an η -neighborhood of the canonical singular strata and above a finite strip cutoff, has total mass one, and satisfies*

$$\|\rho_\eta\|_{\mathcal{A}_{CM}^{\text{adm}}} \leq CM, \quad |\langle F, \rho - \rho_\eta \rangle| \leq \eta \|F\|_{\text{test}}$$

for every branchwise test observable F appearing in the HJB residual ledger. The constant C is independent of ρ and η .

Proof. First remove high homogeneity strips, choosing the cutoff so that the admissible strip tail contributes at most $\eta/3$. Next remove an η -thin neighborhood of the finite singular-stratum family below the cutoff. Its contribution is controlled by the weak trace bound used in Lemma F.5. On the remaining finite family of homogeneous rectangles, mollify in collision coordinates along stable and unstable chart directions and use a C^{r-6} partition of unity. The anisotropic density norm is stable under this chartwise smoothing, up to a constant depending only on the finite atlas. Renormalization changes pairings by $O(\eta)$, because the removed mass is $O(\eta)$ uniformly on the admissible ball. This proves the displayed bounds. $\square$

Proposition G.2 (Perturbed-test smoothing). *Let Φ^ε be the normalized contact-frozen perturbed test used in Lemma E.1. For every compact contact-jet window K , admissible-density radius M , and $\eta > 0$, there is a branchwise C^{r-8} perturbed test Φ_η^ε , supported away from a finite singular-neighborhood and using the same frozen macroscopic jet, such that*

$$\sup_{\rho \in \mathcal{A}_M^{\text{adm}}} |\langle \mathcal{L}_\varepsilon \Phi^\varepsilon - \mathcal{L}_\varepsilon \Phi_\eta^\varepsilon, \rho \rangle| \leq \eta(1 + \|\varphi\|_{C^5})$$

for all sufficiently small ε . The subsolution and supersolution inequalities are unchanged after sending $\eta \downarrow 0$.

Proof. The correctors in Φ^ε are reduced-resolvent images of branchwise finite-response observables and moving-trace letters. Approximate each observable by the chartwise smoothing procedure in Lemma G.1; apply the reduced resolvent to the smoothed zero-mean observable; and subtract the invariant mean to preserve the normalized gauge. Differentiated resolvent words commute with this approximation up to the same weak error, because their letters are bounded by the response package and because the singular-neighborhood contribution is already a trace term. The prelimit operator $\mathcal{L}_\varepsilon$ contains only finitely many branch derivatives on the chosen order window, so the weak pairing error is bounded by the displayed η -term. The viscosity inequalities are first proved for the smoothed tests and then recovered by letting $\eta \rightarrow 0$. $\square$

Corollary G.3 (Smooth approximation closure). *The detailed HJB closure theorem is unchanged if all admissible densities and all microscopic correctors used in the perturbed-test argument are first approximated by branchwise smooth objects supported away from singular strata, with the approximation error sent to zero before the half-relaxed limit.*

Proof. Combine Lemma G.1 and Proposition G.2. The order of limits is: choose a compact contact window, choose the smoothing and strip cutoff so the weak approximation error is below η , send $\varepsilon \rightarrow 0$ in the smoothed prelimit inequality, and finally send $\eta \rightarrow 0$. The branch residual modulus of Proposition F.7 is uniform in this procedure. $\square$

APPENDIX H. HALF-RELAXED LIMITS

Lemma H.1 (Contact selection for normalized gauges). *Let $\varphi \in C^3$ touch the upper half-relaxed paired limit strictly at (t_0, x_0) . Then there are prelimit contact points $(t_\varepsilon, x_\varepsilon) \rightarrow (t_0, x_0)$ and admissible densities ρ_ε such that the normalized contact-frozen perturbed test is an upper viscosity-duality test for u^ε . The lower-contact statement holds with inequalities reversed.*

Proof. Add a small strict macroscopic bump and work on a compact cylinder. The paired half-relaxed definition gives approximate maximizing triples. The microscopic gauge is normalized by subtracting its value at the chosen contact point; this does not alter generator or horizontal derivatives, but it restores the exact contact equality in the paired test. Homogeneous branch approximations avoid the zero-measure iterated singular strata, and high-strip tails are summable by the response-theory trace estimates. A deterministic bump turns approximate contacts into exact dual contacts. The lower-contact case is identical. $\square$

Proposition H.2 (Subsolution and supersolution passage). *The upper half-relaxed paired envelope is a viscosity subsolution of*

$$\partial_t u + \mathcal{H}(x, Du, D^2 u) = 0,$$

and the lower half-relaxed paired envelope is a viscosity supersolution. The same first and second cell equations are used in both halves; only the prelimit viscosity inequality changes sign.

Proof. At an upper contact, use Lemma H.1 and insert the normalized perturbed test in the prelimit subsolution inequality. By Lemma E.1,

$$\partial_t \varphi(t_\varepsilon, x_\varepsilon) + \mathcal{H}(x_\varepsilon, D\varphi, D^2 \varphi) \leq |\langle R_\varepsilon[\varphi], \rho_\varepsilon \rangle|.$$

The residual bound tends to zero on compact contact-jet windows, giving the subsolution inequality. The lower contact gives the reverse inequality for the lower envelope by the same argument. $\square$

Lemma H.3 (Density-to-state upgrade). *Suppose $F_\varepsilon(z)$ is uniformly bounded on homogeneous rectangles, its pairings against every fixed admissible density ball tend to zero, and its oscillation on nested homogeneous subrectangles is bounded by*

$$\text{osc}_{R'} F_\varepsilon \leq \omega_R(\text{diam } R') + r_\varepsilon, \quad r_\varepsilon \rightarrow 0.$$

Then $F_\varepsilon(z) \rightarrow 0$ for Liouville-a.e. microscopic state after discarding the canonical iterated singular set.

Proof. Disintegrate Liouville measure into a countable algebra of homogeneous rectangles. A stable-smoothed density equal to the normalized indicator of a fixed rectangle, up to an arbitrarily small boundary layer, has a finite anisotropic norm. Hence the pairing hypothesis applies to that rectangle. If a positive-measure set had $\limsup |F_\varepsilon| > \eta$, the leafwise differentiation basis and the oscillation modulus would give a smaller fixed rectangle with an average of one sign bounded away from zero, contradicting the pairing limit. Countability gives the almost-everywhere statement. $\square$

Theorem H.4 (Detailed HJB closure). *The prelimit deterministic viscosity-duality solutions converge, after anisotropic microscopic pairing and then almost everywhere in the homogeneous state variable, to the unique viscosity solution of the effective HJB equation. The proof uses only the finite-response port, the imported Paper 1 response theorem, the corrector hierarchy above, and macroscopic viscosity comparison.*

Proof. The upper and lower paired envelopes inherit terminal data from the prelimit terminal condition. By Proposition H.2 they are respectively sub- and supersolutions of the effective HJB. The comparison modulus recorded in the main article gives comparison, so the envelopes coincide with the unique viscosity solution. Uniformity of Lemma E.1 over bounded admissible density balls gives paired convergence. Applying Lemma H.3 on a countable dense family of macroscopic points and then using deterministic macroscopic equicontinuity gives the Liouville-a.e. state statement. $\square$

APPENDIX I. COEFFICIENT AND NONCONVEXITY CHECKS

Proposition I.1 (Coefficient differentiability ledger). *On every compact contact-jet window, the maps*

$$(x, p) \mapsto D(x, p), \quad (x, p) \mapsto H(x, p)$$

have the derivatives used in the comparison and residual estimates. Each derivative is a finite sum of invariant projector derivatives, reduced resolvent derivatives, primitive read-out derivatives, and moving-boundary trace insertions.

Proof. Differentiate the corrector identity

$$\chi_i = -\mathcal{A}_{\Theta(x,p)}^{-1} B_i(x, p, \cdot)$$

in the moving Banach bundle. The derivative of the inverse generator is the usual resolvent word with one differentiated generator letter; higher derivatives give ordered resolvent words. Paper 1 bounds those words, including trace letters caused by moving singular boundaries. Since the finite-response read-outs have bounded branchwise derivatives on the compact window, differentiating the invariant pairings defining D and H gives the stated finite expansion. $\square$

Definition I.2 (Correlation Hilbert bundle). For each compact contact window and each (x, p) , define a bilinear correlation form on centered finite-response observables by

$$\mathfrak{c}_{x,p}(f, g) = \frac{1}{2} \int_0^\infty \int (f(z)g(\Phi_{\Theta(x,p)}^t z) + g(z)f(\Phi_{\Theta(x,p)}^t z)) d\mu_{\Theta(x,p)}(z) dt.$$

The integral is interpreted through the reduced observable resolvent. Its null space is factored out, and the completion is denoted $\mathfrak{H}_{x,p}$. The drift factor is the map

$$\Sigma(x, p)^* e_i = [B_i(x, p, \cdot)]_{\mathfrak{H}_{x,p}}, \quad D(x, p) = \Sigma(x, p) \Sigma(x, p)^*.$$

Lemma I.3 (Green-Kubo positivity and coboundary kernel). *The tensor $D(x, p)$ is symmetric positive semidefinite. Moreover,*

$$a^T D(x, p) a = 0$$

if and only if the centered observable $a \cdot B(x, p, \cdot)$ is a deterministic generator coboundary,

$$a \cdot B(x, p, \cdot) = -\mathcal{A}_{\Theta(x,p)} h$$

for some zero-mean observable h in the reduced generator domain.

Proof. Symmetry is built into $\mathfrak{c}_{x,p}$. Positivity follows from the finite-time variance identity

$$a^T D(x, p) a = \lim_{T \rightarrow \infty} \frac{1}{2T} \int \left(\int_0^T a \cdot B(x, p, \Phi^t z) dt \right)^2 d\mu(z),$$

where convergence is justified by deterministic correlation summability. If $a \cdot B = -\mathcal{A}h$, the time integral telescopes to $h(\Phi^T z) - h(z)$, so the normalized variance vanishes. Conversely, if the variance vanishes, the reduced resolvent $h = -\mathcal{A}^{-1}(a \cdot B)$ has zero correlation energy; the zero-mean spectral gap identifies the observable with the generator range. This is an anisotropic observable-domain statement, not an L^2 mixing assumption. $\square$

Proposition I.4 (Correlation-factor transport). *The fibers $\mathfrak{H}_{x,p}$ in Definition I.2 admit deterministic metric transports $J_{y \rightarrow x}^p : \mathfrak{H}_{y,p} \rightarrow \mathfrak{H}_{x,p}$ on compact coordinate windows such that*

$$\| \Sigma(x, p) - J_{y \rightarrow x}^p \Sigma(y, p) \|_{\text{HS}} \leq C_K |x - y|.$$

This supplies the Crandall-Ishii diffusion factor without taking a matrix square root of a possibly degenerate covariance matrix.

Proof. Differentiate the correlation form in the moving Banach bundle. The derivative is a finite sum of invariant-projector, reduced-resolvent, primitive observable, and moving-boundary trace terms, all bounded by the response package. On the quotient by the null space, Riesz representation gives the unique finite-rank correction that makes the covariant derivative metric:

$$\partial_{x_a} \mathfrak{c}_{x,p}(f, g) = \mathfrak{c}_{x,p}(\nabla_a f, g) + \mathfrak{c}_{x,p}(f, \nabla_a g).$$

Solving the associated linear ODE along the segment from y to x gives the isometric transport $J_{y \rightarrow x}^p$. Since B_i is a finite-response observable with bounded moving-bundle derivative, its transported class is Lipschitz in the correlation Hilbert norm. Summing over the finitely many slow coordinates gives the displayed Hilbert-Schmidt bound. $\square$

Proposition I.5 (Green-Kubo comparison trace estimate). *If $p = \alpha(x - y)$ and X, Y satisfy the Crandall-Ishii block inequality, then on compact windows*

$$-\text{Tr}(D(x, p)X - D(y, p)Y) \leq C\alpha|x - y|^2 + o_{\text{fr}}(1),$$

where $o_{\text{fr}}(1)$ is the finite-rank correlation-bundle tail removed after the Hilbert projection cutoff.

Proof. Transport $\Sigma(y, p)$ into $\mathfrak{H}_{x,p}$ by Proposition I.4. Apply the Ishii matrix inequality to the paired vectors $(\Sigma(x, p)^*\xi, J_{y \rightarrow x}^p \Sigma(y, p)^*\xi)$ in a finite-dimensional subspace of $\mathfrak{H}_{x,p}$. The block inequality gives

$$-\text{Tr}(D(x, p)X - D(y, p)Y) \leq 3\alpha\|\Sigma(x, p) - J_{y \rightarrow x}^p \Sigma(y, p)\|_{\text{HS}}^2 + o_{\text{fr}}(1).$$

The factor-transport bound gives the stated estimate, and the projection tail vanishes by monotone convergence of the positive correlation form and the uniform reduced-resolvent bound. $\square$

Proposition I.6 (Comparison modulus ledger). *The derived operator*

$$\mathcal{H}(x, p, X) = -\text{Tr}(D(x, p)X) - H(x, p)$$

is continuous, degenerate elliptic, and satisfies the Crandall-Ishii modulus on compact contact-jet windows.

Proof. Degenerate ellipticity follows from positivity of the Green-Kubo tensor. The differentiability ledger gives local continuity and Lipschitz control in (x, p) . The Green-Kubo representation gives a Hilbert-factorization $D = \Sigma\Sigma^*$. Applying the Ishii block inequality to $(\Sigma(x, p)^*\xi, \Sigma(y, p)^*\xi)$ and using finite-rank approximation in the correlation Hilbert bundle yields the standard trace estimate. Combining that estimate with the local modulus of H gives the comparison condition. $\square$

Lemma I.7 (Synchronized density for doubled contacts). *In the doubled-variable comparison argument, the same microscopic admissible density can be used in the upper and lower viscosity inequalities after transport through the two Banach-bundle charts. If the two contact fibers are $\Theta(x, p_U)$ and $\Theta(y, p_V)$, then the transported density satisfies*

$$\|\rho_U - \mathfrak{T}_{\Theta(x, p_U) \leftarrow \Theta(y, p_V)} \rho_V\|_w \leq C_M(|x - y| + \|p_U - p_V\|_w) + o_{\text{strip}}(1),$$

where $o_{\text{strip}}(1)$ is sent to zero by the common homogeneity cutoff.

Proof. The normal-coordinate transport between nearby billiard fibers preserves positivity and total mass and has uniformly bounded anisotropic norm on compact contact windows. Interior generator, drift, and potential terms differ by the finite-response Lipschitz modulus. Physical collision faces have cancelling incoming and outgoing flux by Lemma C.5. Grazing and branch-change discrepancies are moving-boundary trace terms, hence are bounded by Proposition C.6. This gives the displayed modulus and permits a single synchronized density in the doubled pairing. $\square$

Proposition I.8 (Doubled-variable comparison refinement). *Let u be a bounded upper semicontinuous subsolution and v a bounded lower semicontinuous supersolution of the derived terminal-value*

HJB, with ordered terminal data. On compact cotangent windows the standard Crandall-Ishii comparison proof closes with the coefficient modulus of Proposition I.6 and the synchronized-density estimate of Lemma I.7. In particular no additional microscopic order assumption is needed.

Proof. Maximize

$$u(t, x) - v(s, y) - \frac{|x - y|^2}{2\alpha} - \frac{|t - s|^2}{2\alpha} - \delta(|x|^2 + |y|^2)$$

on the compact doubled torus. The terminal ordering puts the maximizing point in the parabolic interior for small parameters. Ishii's lemma supplies common gradient $p_\alpha = (x - y)/\alpha$ and matrices X, Y satisfying the usual block inequality. The diffusion term is estimated by applying that block inequality to a finite-rank approximation of the Green-Kubo correlation factor. The error is then removed in the correlation Hilbert bundle. The Hamiltonian part is controlled by the finite-response modulus in x and p .

For the prelimit paired comparison used in the half-relaxed proof, the upper and lower microscopic tests are first transported to one reference fiber. Lemma I.7 bounds the density-transport error, while Proposition C.6 removes the singular trace tail. Subtracting the two viscosity inequalities leaves

$$0 \leq o_\alpha(1) + o_\delta(1)$$

at a hypothetical positive maximum, a contradiction after sending $\alpha \downarrow 0$ and then $\delta \downarrow 0$. Perron stability and terminal-data Lipschitz dependence follow by the same comparison with constant barriers, using the zero-energy gauge $H(x, 0) = 0$. $\square$

Proposition I.9 (Terminal-value stability and zero-energy gauge). *The effective terminal-value HJB has a unique bounded uniformly continuous viscosity solution and the solution operator $S_t\phi = u(0, \cdot)$ satisfies*

$$\|S_t\phi - S_t\psi\|_\infty \leq \|\phi - \psi\|_\infty, \quad S_t(\phi + c) = S_t\phi + c.$$

The same statements hold for the paired prelimit problem on each fixed $\varepsilon > 0$ in the admissible-density order.

Proof. Comparison from Proposition I.8 gives uniqueness. For stability, compare the solution with terminal datum ϕ to the solutions with terminal data $\psi + \|\phi - \psi\|_\infty$ and $\psi - \|\phi - \psi\|_\infty$. Constants solve the homogeneous equation because the zero-energy normalization gives $H(x, 0) = 0$ and the diffusion term vanishes on constants. The prelimit paired statement is identical: the branchwise Hamilton principal function is shifted by the terminal constant, the microscopic generator kills constants, and the paired order is separated by admissible densities. $\square$

Proposition I.10 (Concrete parameter window for the two-mode port). *For the triangular Lorentz cell and the two disjoint deformation bumps used in the main article, there is a nonempty open parameter window in which the finite-horizon geometry remains unchanged and the nonconvexity dominance condition holds. Let $\theta_* < \theta_{\text{geom}}$ be small enough to preserve curvature, separation, and corridor blocking for all normal displacements with C^{r+3} -norm at most θ_* . Choose the action coefficient $\lambda(x)$, quartic stabilizer $\kappa(x) > 0$, and response amplitudes so that*

$$2\lambda(x)\langle \eta \rangle_{\theta_0} > C_{\text{resp}}(x) + 2\delta_x, \quad \|\Theta(x, p) - \theta_0\|_{C^{r+3}} \leq \theta_*/2$$

on the working cotangent window. Then every sufficiently small perturbation of the port coefficients inside this window still satisfies the strict dominance inequality with margin at least δ_x .

Proof. The finite-horizon triangular Lorentz inequalities are open in the C^{r+3} topology of the scatterer boundary, so a geometric radius $\theta_{\text{geom}} > 0$ exists. The normalized deformation bumps and the observable bump have disjoint supports, hence small amplitudes keep the table in the same finite-horizon class and avoid branch interference among the two response modes.

The response remainder $C_{\text{resp}}(x)$ is continuous in the primitive read-out norms and in the Paper 1 response constants over the compact working window. The observable η is nonnegative and

nonzero on a regular collision arc, so $\langle \eta \rangle_{\theta_0} > 0$. Increasing $\lambda(x)$ within the finite-response energy budget, or first reducing the response amplitudes and then fixing λ , gives the strict margin $2\delta_x$. Continuity leaves margin δ_x under nearby parameter choices. The positive quartic term keeps the example inside the compact cotangent window and does not remove the negative second directional coefficient at the origin. $\square$

Corollary I.11 (Concrete parameter-choice recipe). *The nonconvex port can be chosen by the following finite sequence.*

- (i) Fix the triangular Lorentz cell and choose the two disjoint normalized boundary bumps and the nonnegative observable bump η .
- (ii) Choose a geometric radius θ_* inside the finite-horizon window, and choose response amplitudes so that $\|\Theta(x, p) - \theta_0\|_{C^{r+3}} \leq \theta_*/2$.
- (iii) Compute the response-budget bound $C_{\text{resp}}(x)$ from the imported Paper 1 constants on the compact cotangent window.
- (iv) Choose $\lambda(x)$ so that $2\lambda(x)\langle \eta \rangle_{\theta_0} > C_{\text{resp}}(x) + 2\delta_x$.
- (v) Choose a positive quartic stabilizer $\kappa(x)$ large enough to keep the action window compact without changing the negative second directional coefficient at $p = 0$.

Every inequality in this list is open; hence nearby smooth parameter choices give the same nonconvexity certificate.

Proof. The first two steps put the table family in the geometric window. The third step is a finite evaluation of already imported response constants. Since $\eta \geq 0$ and is not identically zero on a regular collision arc, its invariant average is positive, so the fourth step is possible. The quartic term has zero second derivative at the origin in the chosen direction after normalization and therefore affects confinement but not the local nonconvex coefficient. Openness follows from continuity of the finite-response read-outs and of the response-budget constants over the compact window. $\square$

Definition I.12 (Dimensionless parameter audit table). For a compact cotangent window normalize the concrete-port quantities by

$$\mathcal{R} = \frac{2\lambda\langle \eta \rangle_{\theta_0}}{C_{\text{resp}} + 2\delta_x}, \quad \mathcal{A} = \frac{\|\Theta - \theta_0\|_{C^{r+3}}}{\theta_*}, \quad \mathcal{S} = \frac{\kappa}{1 + \lambda + \|\Theta - \theta_0\|_{C^{r+3}}}.$$

The following dimensionless choices are sufficient for the port used in the main paper:

Quantity	Audit value	Consequence
geometric amplitude ratio $\mathcal{A}$	$\leq 1/2$	the table stays in the finite-horizon billiard window.
nonconvex dominance ratio $\mathcal{R}$	≥ 2	the negative direct action term dominates all response-corrector terms.
quartic stabilizer ratio $\mathcal{S}$	≥ 4	the action remains confined on the selected cotangent window.
smoothing radius δ_{sm}	$\leq 1/16$ of the bump separation	the two boundary bumps and the observable bump remain disjoint.
strip-tail tolerance	$\leq 10^{-3}\delta_x$	the high-strip contribution cannot close the nonconvexity gap.

This is an audit table for normalized inequalities, not a numerical simulation of a billiard orbit.

Proposition I.13 (Parameter audit verification). *If the dimensionless inequalities in Definition I.12 hold on the compact cotangent window, then the concrete two-mode port satisfies the hypotheses of Proposition I.10 and Corollary I.11 after the smoothing and strip cutoffs of Corollary G.3.*

Proof. The amplitude bound $\mathcal{A} \leq 1/2$ leaves half of the finite-horizon geometric margin unused, so the smoothed port cannot create a new grazing family. The dominance bound $\mathcal{R} \geq 2$ gives

$$2\lambda\langle\eta\rangle_{\theta_0} \geq 2(C_{\text{resp}} + 2\delta_x),$$

so even after losing C_{resp} to response-corrector terms and δ_x to smoothing, a gap of at least δ_x remains. The stabilizer bound keeps the finite-response action coercive on the chosen cotangent window and does not change the second derivative at the origin in the nonconvex direction. The bump-separation condition preserves the sign and support assumptions used in the direct action computation. Finally, the strip-tail tolerance is smaller than the remaining gap, so Proposition F.6 cannot erase the certificate. Thus all inequalities in the concrete parameter recipe remain valid after the regularization steps. $\square$

Definition I.14 (Normalized numerical certificate). The following dimensionless values give an inspectable certificate for the concrete-port inequalities. Work in units in which

$$C_{\text{resp}} = 1, \quad \langle\eta\rangle_{\theta_0} = 1, \quad \theta_* = 1, \quad b_{\text{sep}} = 1,$$

where b_{sep} is the minimum separation among the two boundary bumps and the observable bump. Choose

$$\delta_x = \frac{1}{10}, \quad \lambda = \frac{3}{2}, \quad \|\Theta - \theta_0\|_{C^{r+3}} = \frac{2}{5}, \quad \kappa = 16, \quad \delta_{\text{sm}} = \frac{1}{32}, \quad \tau_{\text{strip}} = 5 \cdot 10^{-5}.$$

Then

$$\mathcal{A} = \frac{2}{5}, \quad \mathcal{R} = \frac{3}{6/5} = \frac{5}{2}, \quad \mathcal{S} = \frac{16}{1 + 3/2 + 2/5} = \frac{160}{29} > 5,$$

and

$$\delta_{\text{sm}} < \frac{b_{\text{sep}}}{16}, \quad \tau_{\text{strip}} < 10^{-3}\delta_x.$$

This is a numerical illustration of the normalized inequalities in Definition I.12; it is not an orbit simulation and it does not estimate the Paper 1 response constants from trajectory data.

Proposition I.15 (Numerical certificate margin). *The certificate in Definition I.14 is stable under small smooth changes of the finite-response port. More precisely, on any compact window where*

$$\mathcal{A} \leq \frac{9}{20}, \quad \mathcal{R} \geq \frac{9}{4}, \quad \mathcal{S} \geq 4, \quad \delta_{\text{sm}} \leq \frac{b_{\text{sep}}}{24}, \quad \tau_{\text{strip}} \leq 8 \cdot 10^{-4}\delta_x,$$

the smoothed two-mode port still satisfies the nonconvexity certificate after the response-corrector, smoothing, and high-strip losses are subtracted.

Proof. The amplitude inequality keeps a strict geometric buffer $1 - \mathcal{A} \geq 11/20$, so the deformation remains inside the finite-horizon window and the bump supports remain separated after smoothing. The stabilizer inequality gives the compact cotangent confinement used in Corollary I.11.

For the nonconvex Taylor coefficient, write the direct contribution as

$$2\lambda\langle\eta\rangle_{\theta_0} = \mathcal{R}(C_{\text{resp}} + 2\delta_x).$$

After subtracting the response-corrector budget, the smoothing loss, and the high-strip tail, the remaining gap is bounded below by

$$(\mathcal{R} - 1)C_{\text{resp}} + (2\mathcal{R} - 1)\delta_x - \tau_{\text{strip}}.$$

With $\mathcal{R} \geq 9/4$ and $\tau_{\text{strip}} \leq 8 \cdot 10^{-4}\delta_x$, this lower bound is positive and in particular exceeds δ_x on the normalized window. Hence the directional second coefficient stays negative. The inequalities are open in the port coefficients because all read-outs and imported response constants are continuous on the compact response window, so nearby smooth ports inherit the same certificate. $\square$

Lemma I.16 (Directional nonconvexity certificate). *Let e be a unit direction and suppose the directional Taylor coefficient*

$$\mathfrak{c}_e(x) = \left. \frac{d^2}{ds^2} H(x, se) \right|_{s=0}$$

is negative on an open set. Then $p \mapsto H(x, p)$ is locally nonconvex there. In the concrete two-mode port of the main paper it is enough that

$$2\lambda(x)\langle\eta\rangle_{\theta_0} > C_{\text{resp}}(x) + \delta_x.$$

Proof. Taylor expansion gives

$$H(x, se) + H(x, -se) - 2H(x, 0) = \mathfrak{c}_e(x)s^2 + o(s^2),$$

which is negative for small nonzero s . For the concrete port, the direct action contribution to $\mathfrak{c}_e$ is $-2\lambda(x)\langle\eta\rangle_{\theta_0}$. The remaining invariant-density and corrector-response terms are bounded by $C_{\text{resp}}(x)$. The dominance inequality therefore forces $\mathfrak{c}_e(x) < 0$. By Proposition I.10, this is realized on an open set of finite-response port coefficients before the transfer-resolvent calculation is performed. $\square$

Lemma I.17 (Generator-level subadditivity obstruction). *Let $S_t\phi = \mathcal{E}_t^\theta[\phi]$. If S_t were subadditive for all smooth localized terminal payoffs, then the Hamiltonian would satisfy*

$$H(x, p + q) \geq H(x, p) + H(x, q)$$

with the terminal-time sign convention of the main article. Hence any strict opposite short-time Hamiltonian defect gives a failure of subadditivity.

Proof. For a smooth payoff with gradient p and zero Hessian at x , the terminal-value HJB gives

$$S_t\phi(x) = \phi(x) - tH(x, p) + o(t).$$

Apply this to two localized affine payoffs with gradients p and q , and to their sum. Subadditivity $S_t(\phi + \psi) \leq S_t\phi + S_t\psi$ implies the displayed inequality after dividing by t and passing $t \downarrow 0$. Therefore a strict reverse defect of the Hamiltonian at the chosen gradients produces the claimed failure. $\square$

Proposition I.18 (Appendix-to-main crosswalk). *The technical blocks above support the following main-paper assertions.*

Main-paper assertion	Appendix support
Finite-response coefficients are microscopic	Definition B.1 and Proposition B.2
Corrector hierarchy and residual identity	Definitions D.6 and F.3, Propositions D.5, D.7, E.2 and F.7, and Lemmas D.2 and E.1
Half-relaxed HJB passage	Lemmas G.1, H.1 and I.7, Propositions G.2, H.2 and I.8, Corollary G.3, and Theorem H.4
Coefficient comparison and differentiability	Propositions I.1, I.4 to I.6 and I.9 and Definition I.2
Nonconvexity and non-subadditivity	Propositions I.10, I.13 and I.15, Corollary I.11, Definitions I.12 and I.14, and Lemmas I.16 and I.17

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